Enabling Dynamic Network Access Control with Anomaly-based IDS and SDN

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Outline

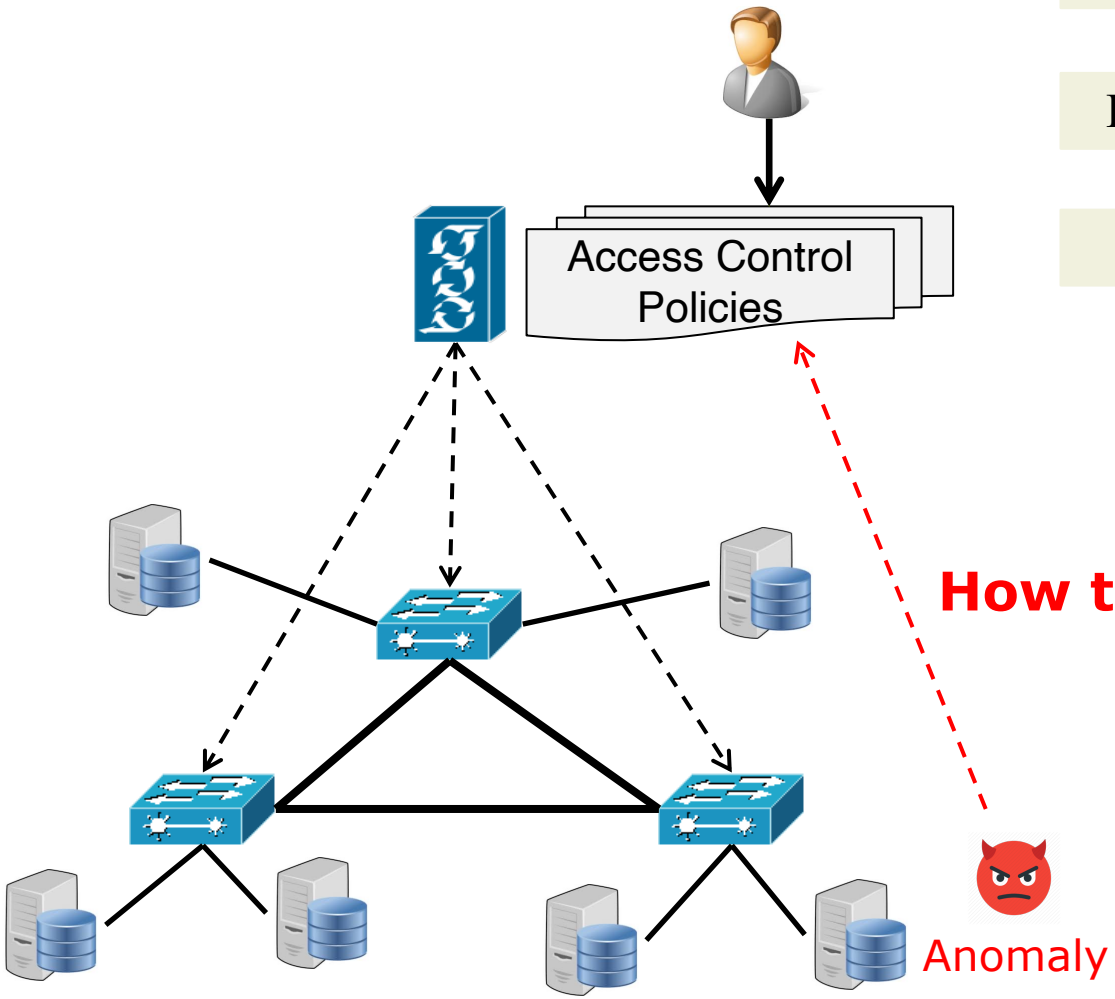
- Motivation
- Background
- Our Approach
- Case Study

Network Access Control with SDN

FlowGuard [HotSDN'14]

Dynamic Firewall [RAID'15]

Virtual Firewall [NDSS'17]

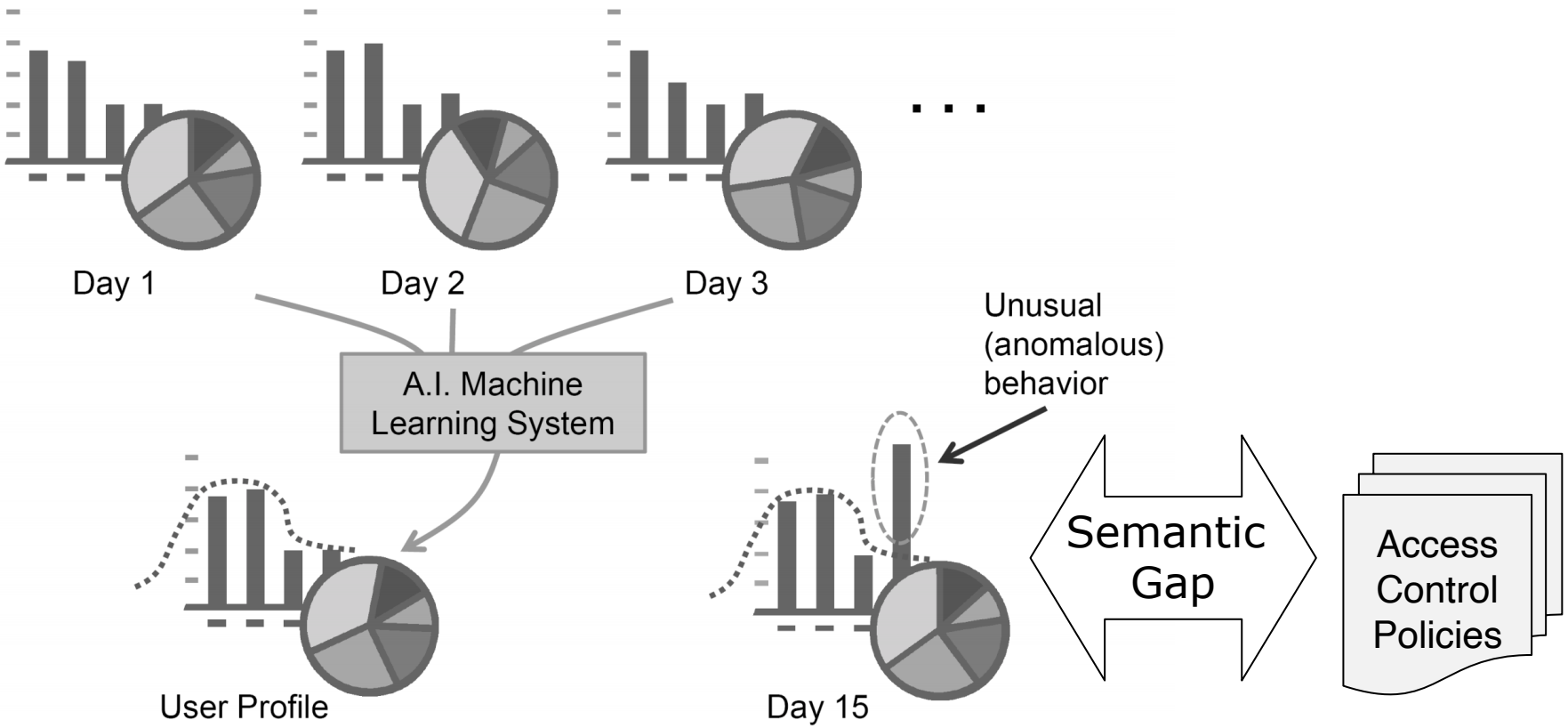


How to generate new ACP?

Unknown vulnerabilities

Zero-day security threat

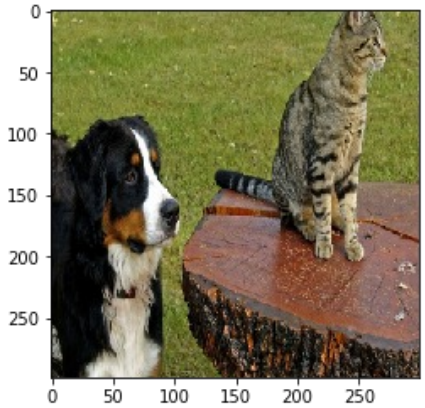
Existing Anomaly-based IDS



Uncover novel security threats

Obscure outcome

Machine Learning Model Explanation



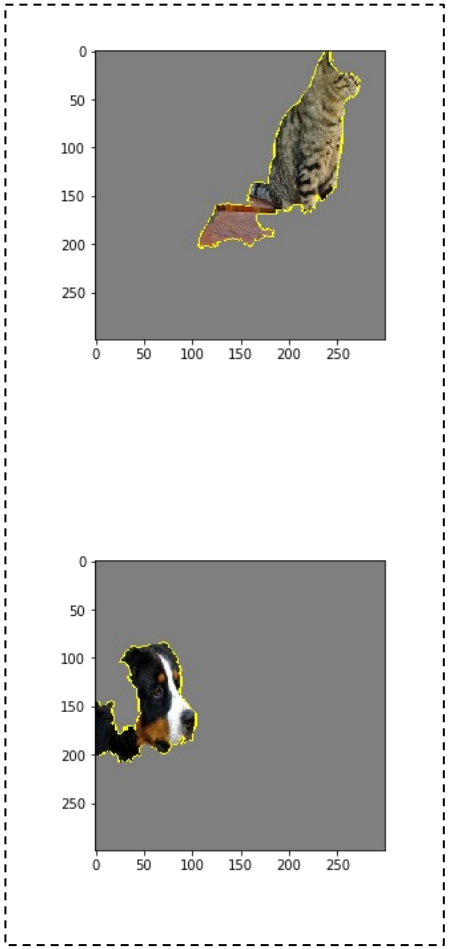
Input

Predictor

CAT

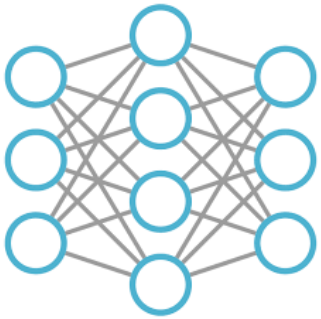
DOG

Outcome

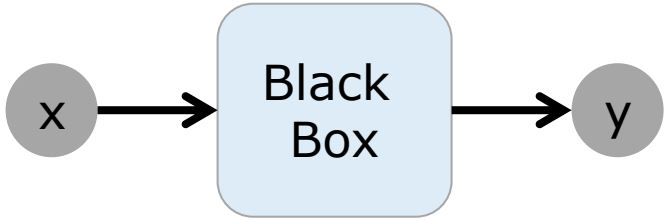


Explanation

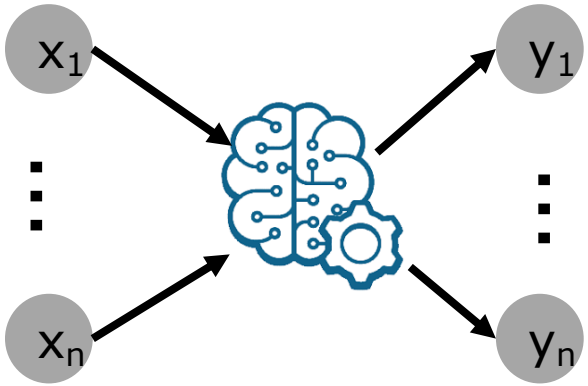
Explanation Mechanisms



Whitebox



Blackbox

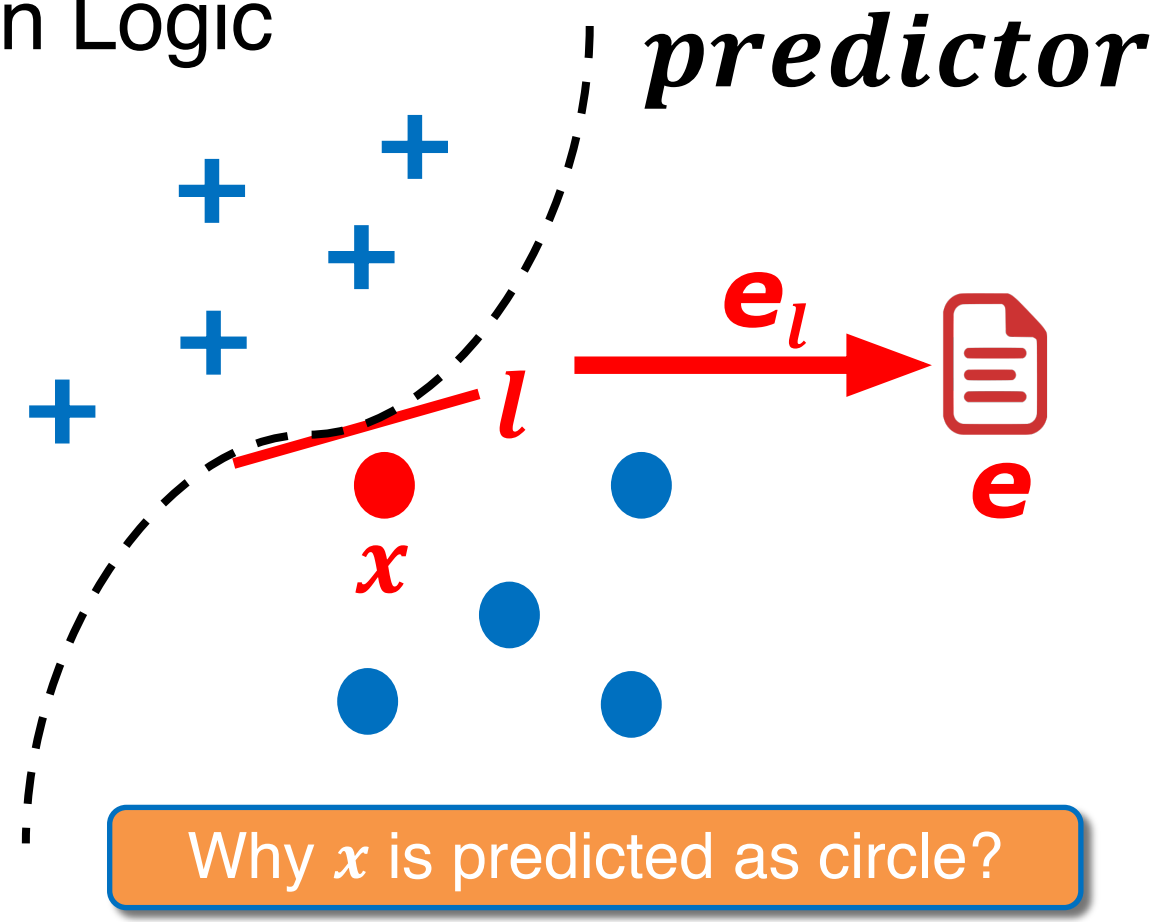


Global Explanation

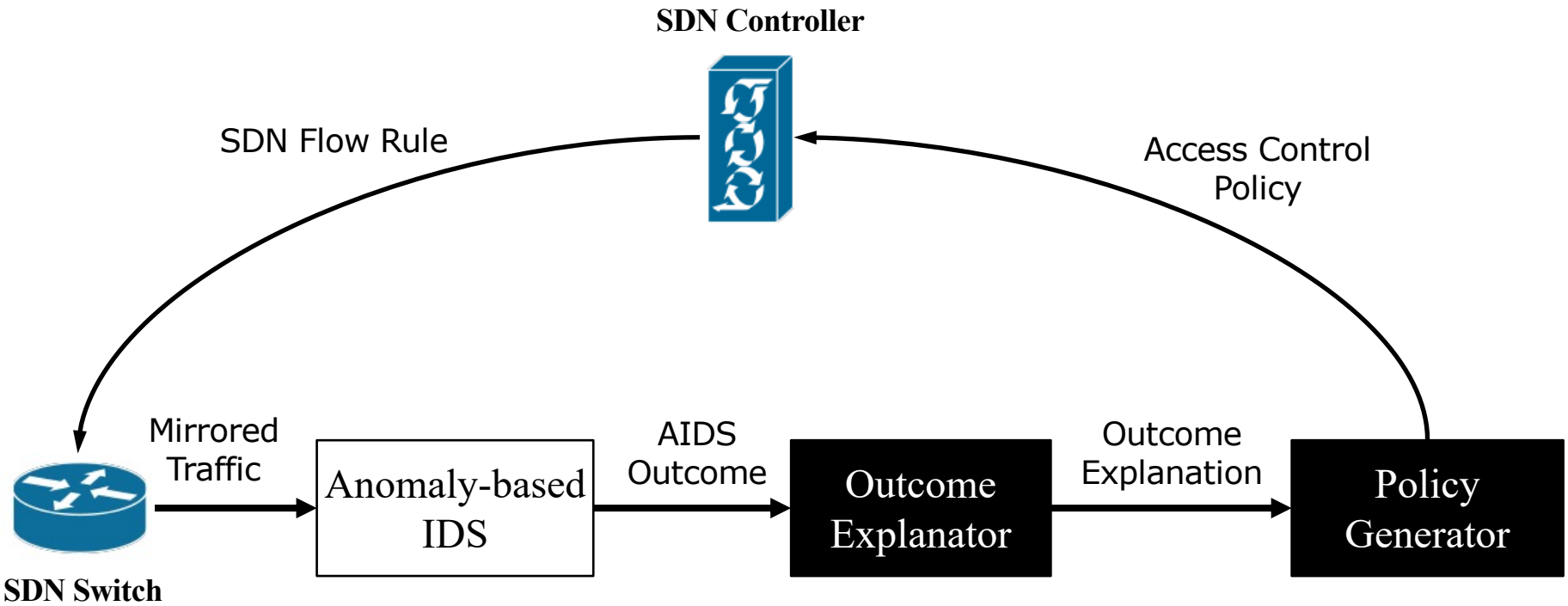
Local Explanation

Local & Blackbox Explanation

1. Local Approximation
2. Explanation Logic



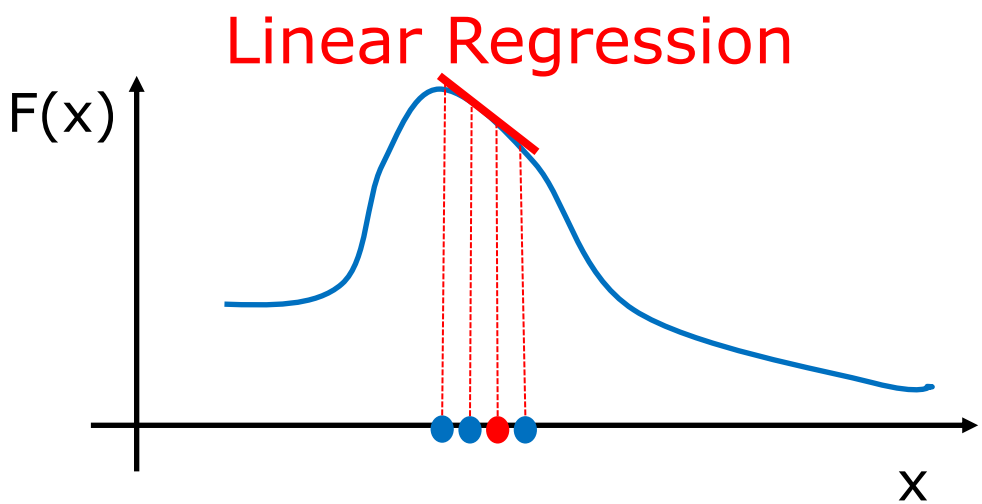
Approach Overview



AIDS Outcome Explanation

1. Local Approximation
2. Explanation Logic

$$l(\mathbf{x}) = \boldsymbol{\alpha}\mathbf{x} + \epsilon$$



$$L(l(\mathbf{x}), y) = \sum_{i=1}^{N} \| \boldsymbol{\alpha}\mathbf{x}_{\boldsymbol{i}} - y_i \|$$

Feature Importance

$\boldsymbol{x}$: (*duration, proto_type, service, flag, src_byte, dst_byte, …*)
FI: (*97, 96, 99, 100, 95, 98, …*)

Access Control Policy Generation

<filers, actions>

Selects network entities

Networks;
Hosts;
Connections;
Flows;
Packets;
Combination of above;

Defines action to take

Allow;
Deny;
Redirect;
Quarantine;
Mirror;
…

$\boldsymbol{x}$: (*duration*, *proto_type*, *service*, *flag*, *src_byte*, *dst_byte*, …)
FI: (*97*, *96*, *99*, *100*, *95*, *94*, …)

Explanation

Case Study: AIDS

- Recurrent Neural Network (RNN)
  - Detect across multiple records
- NSL-KDD dataset
  - 41 raw feature
- Keras + TensorFlow for implementation

Case Study: Outcome Explanation

- Choose *Neptune attack* in dataset
  - Extensive SYN error or SYN rejection
- Two records labeled as Neptune attack

Record1: (0, tcp, private, S0, …, 255, 20, 0.08, 0.07, 0, 0, 1, 1, 0, 0)

Record2: (0, tcp, imap4, REJ, …, 255, 17, 0.07, 0.07, 0, 0, 0, 0, 1, 1)

- Explanation (Feature Importance)

Record	Duration	Proto.	Service	Flag	…	Dst SYN E.	Dst SYN Srv.E.	Dst Rej E.	Dst Rej Srv. E.
Record1	*0*	*tcp*	*private*	*S0*	…	*1*	*1*	*0*	*0*
Record2	*0*	*tcp*	*imap4*	*REJ*	…	*0*	*0*	*1*	*1*

Percentage of SYN Error

Percentage of Rejection Error

Case Study: Policy Generation

Outcome Explanation

Record	Duration	Proto.	Service	Flag	...	Dst SYN E.	Dst SYN Srv.E.	Dst Rej E.	Dst Rej Srv. E.
Record1	*0*	*tcp*	*private*	*S0*	...	*1*	*1*	*0*	*0*
Record2	*0*	*tcp*	*imap4*	*REJ*	...	*0*	*0*	*1*	*1*

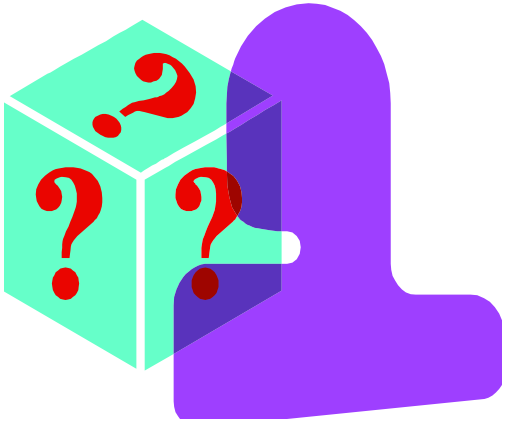
<filters=(*ip_proto=tcp***, *tcp_flags=syn*, *sip=192.168.1.2, dip=192.168.1.3*),**
actions=(*drop***)>**

Access Control Policy

Conclusion and Future Work

- Conclusion
  - Explained the outcome of anomaly-based IDS
  - Generated network access control policy according to the explanation
- Future work
  - Better explanation that handles decency among records
  - Policy generation process formalization
  - More evaluation on realistic traffic and attacks

Q & A



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Thank you!